Shared vs. Independent Stratification Vector (f) Under Real LD Structure

A cross-scale, cross-chromosome comparison using 1000 Genomes chr1 and chr22 LD structure

Executive Summary

This analysis extends the project's AR1-based heritability simulation framework to use **real linkage-disequilibrium (LD) structure** drawn from 1000 Genomes chromosome 1 and chromosome 22 genotypes (in place of synthetic AR1 correlation), and tests how a reference panel's stratification vector f behaves when it is **shared** with the main sample versus drawn **independently**. The comparison was run across five sample/SNP scales on chr1 (n=500/m=1000, n=1000/m=2000, n=2000/m=4000, n=5000/m=10000, n=10000/m=20000) and four scales on chr22 (n=500 through n=5000 - chr22 has not yet been run at n=10000), for a total of 848 simulated datasets (100 replicates each) across all scenarios, scales and chromosomes.

Headline finding: without a reference panel, whether f is shared or redrawn per replicate makes almost no difference to the LDSC free-intercept h2 estimate across every scale tested. **With a reference panel**, however, an f-mismatched reference panel produces estimates that increasingly diverge from the shared-f case as F_ST grows - and this divergence gets **substantially worse** as sample size and SNP count scale up, not better (see Table 1 for n=500-5000; F_ST=0.3/0.5 were not re-run at n=10000, but the same widening pattern is already visible within the 0-0.1 range shown in the n=10000 figures below).

This is a different failure mode than was seen in the project's earlier AR1-based reference-panel work: there, an f-mismatched reference panel caused the free-intercept estimator to *collapse toward zero*. Under real LD structure, the same mismatch instead causes the estimator to *inflate*, and the inflation grows with scale.

Methods

Genotypes were simulated with LD structure taken from real 1000 Genomes EUR chr1 (106,136 SNPs) and chr22 (18,613 SNPs) PLINK files, using the data_generation class's realistic=True path: synthetic individuals are drawn as random linear combinations of the real genotype matrix's columns, preserving its LD structure while allowing arbitrary sample size. A genetic-drift/stratification vector f (parameterized by F_ST) splits the sample into two subpopulations, exactly as in the project's AR1 pipeline.

Four scenarios were run at every scale and chromosome (chr22 through n=5000; chr1 additionally at n=10000):

- **No ref panel, shared f** - LD scores computed in-sample; f drawn once per F_ST and reused across all 100 replicates.
- **No ref panel, independent f** - LD scores computed in-sample; f redrawn fresh every replicate.
- **Ref panel, shared f** - LD scores/GWASH statistics taken from a separately-simulated reference panel; the reference panel's f is shared with the main replicates.
- **Ref panel, independent f** - same reference panel LD scores, but the main replicates draw their own f, independent of (mismatched with) the reference panel's.

The reported estimator is LDSC with a free intercept (h2_ldsc_reg), averaged over 100 replicates per F_ST value. The figures below focus on F_ST in [0, 0.1] (F_ST in {0, 0.001, ..., 0.01, 0.02, ..., 0.1}); Table 1 further below still reports F_ST=0.3/0.5 for the scales where that older, wider grid was run (n=500 through n=5000), for reference against earlier findings - n=10000 only has the 0-0.1 grid and so is omitted from Table 1. A theoretical reference line h2_pop + σ_s^2/C_f (C_f = mean(f^2/(1+f^2))) is overlaid on each plot.

Fixed parameters throughout: h2_pop=0.3, sigma_s=0.3, seed=123. m is always 2n.

Summary: h2 estimate gap (independent f – shared f) at F_ST=0.5

Mean LDSC free-intercept h2 estimate over 100 replicates at F_ST=0.3 and F_ST=0.5, for shared-f and independent-f, in both the no-ref-panel and ref-panel settings. n=10000 is not shown here - that scale was only run on the newer 0-0.1 F_ST grid.

Chrom	n	m	Setting	Shared f (F_ST=0.3)	Indep f (F_ST=0.3)	Shared f (F_ST=0.5)	Indep f (F_ST=0.5)	Gap (F_ST=0.5)
chr1	500	1000	no ref panel	0.514	0.490	0.447	0.422	-0.025
chr1	500	1000	ref panel	0.541	0.605	0.508	0.869	+0.362
chr1	1000	2000	no ref panel	0.478	0.489	0.428	0.422	-0.006
chr1	1000	2000	ref panel	0.526	1.052	0.412	1.512	+1.100
chr1	2000	4000	no ref panel	0.451	0.427	0.396	0.377	-0.020
chr1	2000	4000	ref panel	0.346	1.726	0.317	2.532	+2.216
chr1	5000	10000	no ref panel	0.396	0.433	0.343	0.386	+0.043
chr1	5000	10000	ref panel	0.515	3.037	0.449	4.529	+4.080
chr22	500	1000	no ref panel	0.568	0.529	0.490	0.470	-0.020
chr22	500	1000	ref panel	0.525	0.656	0.478	0.967	+0.490
chr22	1000	2000	no ref panel	0.485	0.450	0.430	0.388	-0.042
chr22	1000	2000	ref panel	0.491	1.026	0.429	1.478	+1.049
chr22	2000	4000	no ref panel	0.454	0.432	0.399	0.382	-0.017
chr22	2000	4000	ref panel	0.349	1.716	0.334	2.481	+2.147
chr22	5000	10000	no ref panel	0.392	0.426	0.341	0.382	+0.041
chr22	5000	10000	ref panel	0.510	2.942	0.442	4.394	+3.952

Summary: h2 estimate at low F_ST (F_ST = 0, 0.005, 0.01)

Mean LDSC free-intercept h2 estimate over 100 replicates at F_ST=0, F_ST=0.005, and F_ST=0.01, for shared-f and independent-f, in both the no-ref-panel and ref-panel settings, across every available scale including the new n=10000/m=20000 chr1 rows at the end.

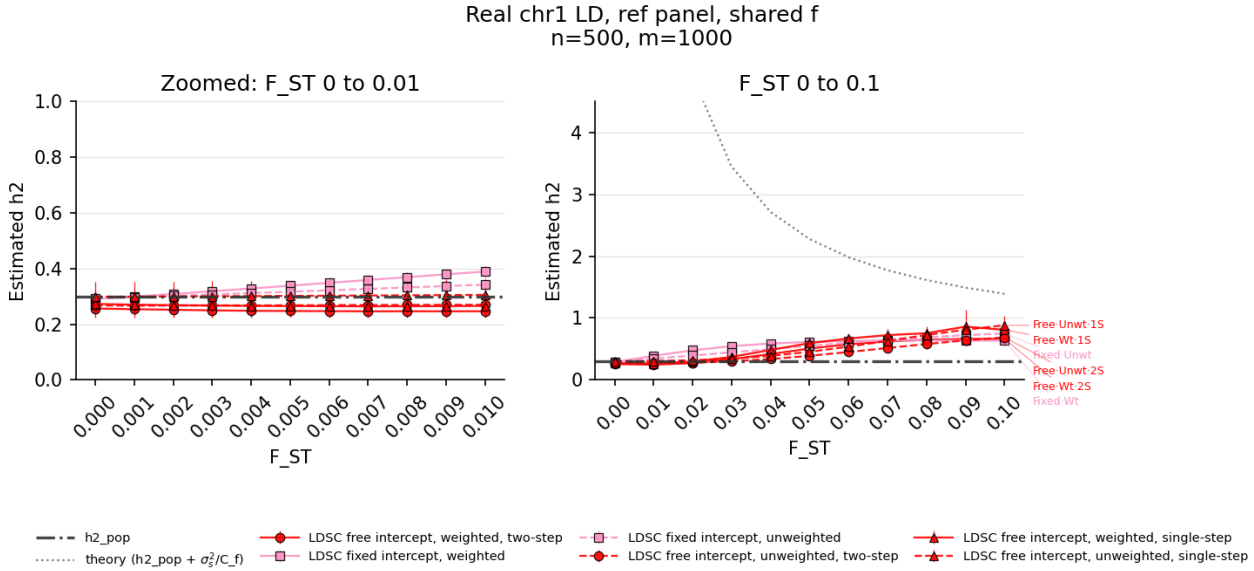
Chrom	n	Setting	Shared f (F_ST=0)	Indep f (F_ST=0)	Shared f (F_ST=0.005)	Indep f (F_ST=0.005)	Shared f (F_ST=0.01)	Indep f (F_ST=0.01)
chr1	500	no ref panel	0.273	0.260	0.261	0.255	0.259	0.259
chr1	500	ref panel	0.256	0.239	0.247	0.233	0.246	0.231
chr1	1000	no ref panel	0.260	0.263	0.250	0.258	0.256	0.270
chr1	1000	ref panel	0.253	0.253	0.237	0.249	0.243	0.246
chr1	2000	no ref panel	0.228	0.247	0.224	0.241	0.249	0.263
chr1	2000	ref panel	0.224	0.242	0.228	0.235	0.261	0.239
chr1	5000	no ref panel	0.195	0.195	0.228	0.229	0.304	0.308
chr1	5000	ref panel	0.192	0.192	0.232	0.223	0.322	0.277
chr1	10000	no ref panel	0.186	0.193	0.276	0.275	0.398	0.390
chr1	10000	ref panel	0.185	0.193	0.279	0.272	0.404	0.372
chr22	500	no ref panel	0.302	0.293	0.300	0.292	0.310	0.300
chr22	500	ref panel	0.303	0.288	0.307	0.287	0.324	0.284
chr22	1000	no ref panel	0.275	0.270	0.274	0.264	0.285	0.274
chr22	1000	ref panel	0.262	0.256	0.256	0.250	0.264	0.247
chr22	2000	no ref panel	0.231	0.237	0.226	0.235	0.246	0.259
chr22	2000	ref panel	0.224	0.229	0.224	0.226	0.251	0.233
chr22	5000	no ref panel	0.196	0.191	0.224	0.226	0.288	0.293
chr22	5000	ref panel	0.195	0.189	0.234	0.224	0.313	0.276

Figures: Chromosome 1

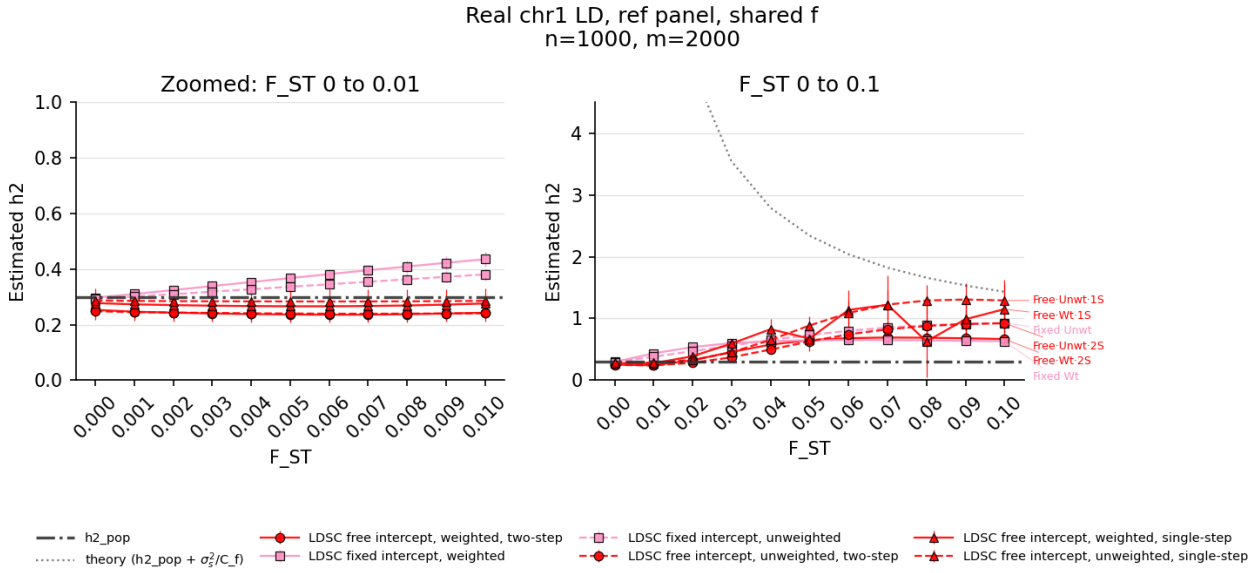
Each figure shows mean h2 (±95% CI) vs. F_ST for all six LDSC estimator variants, within one (reference-panel setting, f-sharing) combination, with the theoretical line overlaid (dotted). Color encodes intercept type (red=free intercept, pink=fixed intercept); linestyle encodes SNP weighting (solid=weighted, dashed=unweighted); marker shape encodes estimation mode (circle=two-step, triangle=single-step, square=fixed-intercept, which has no re-estimation step). Left panel zooms into F_ST 0-0.01; right panel shows F_ST 0-0.1.

Reference-Panel Setting (the main finding)

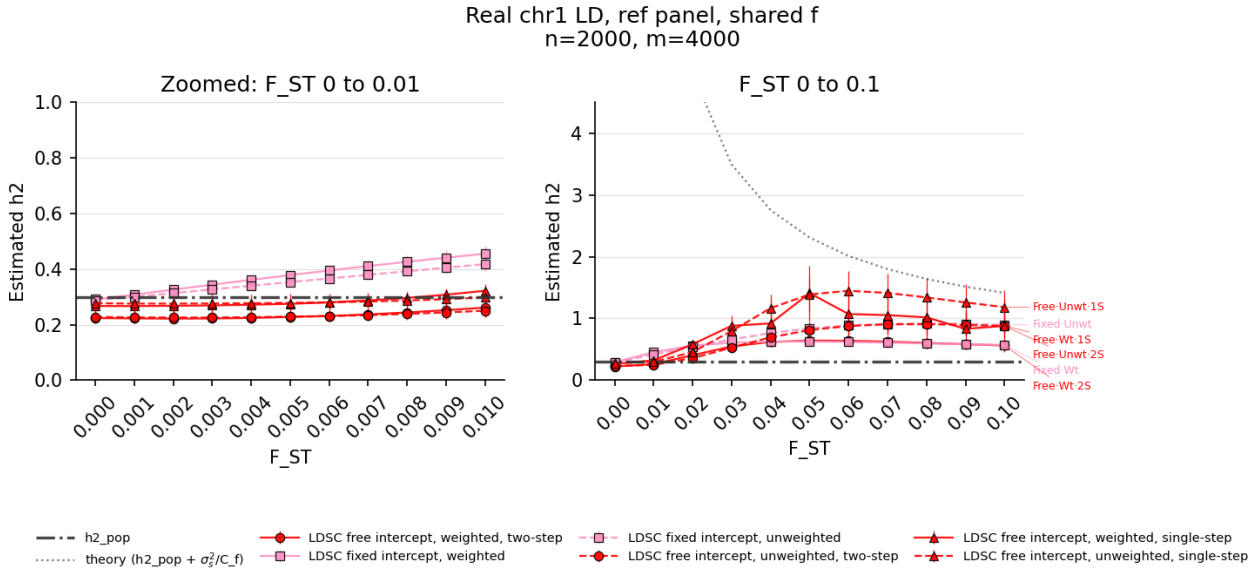
ref panel, shared f



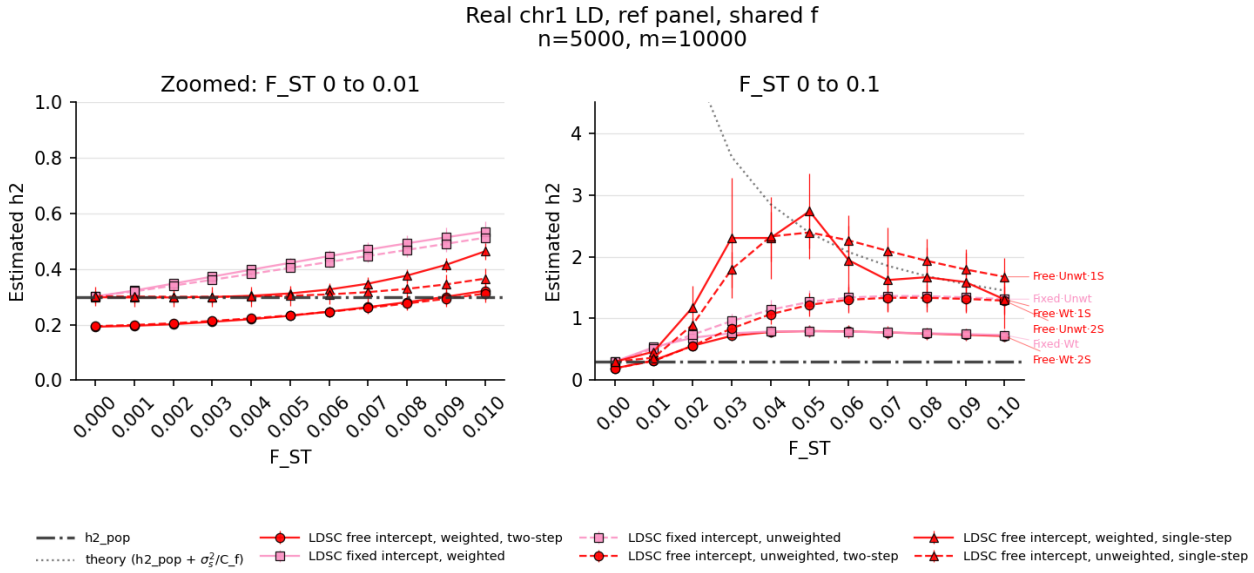
chr1, n=500, m=1000 — ref panel, shared f



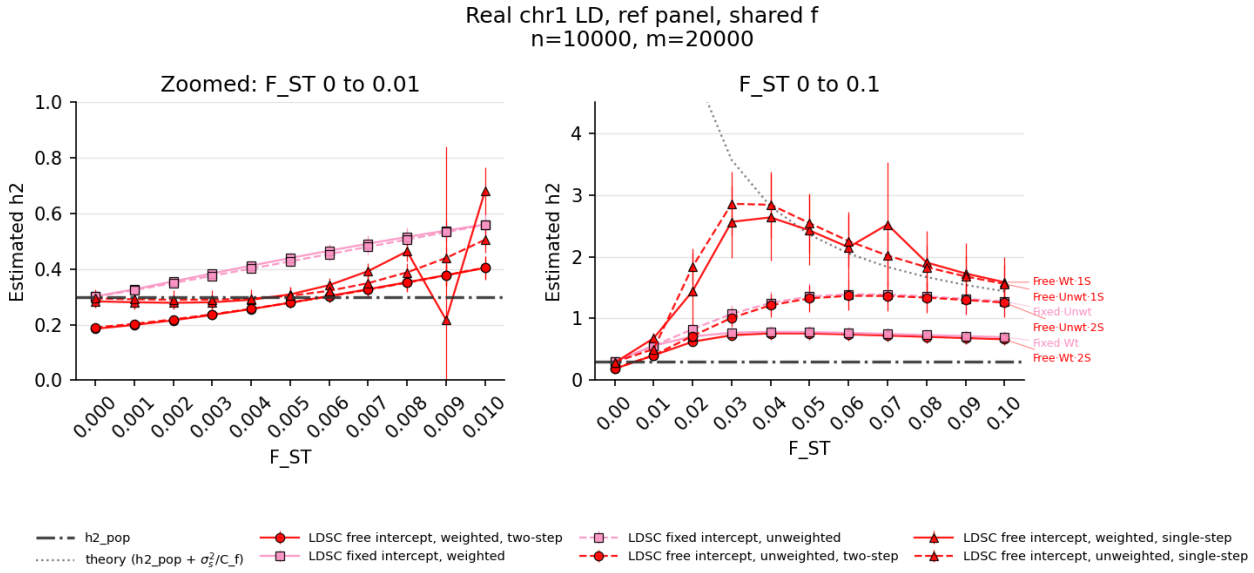
chr1, n=1000, m=2000 — ref panel, shared f



chr1, n=2000, m=4000 — ref panel, shared f

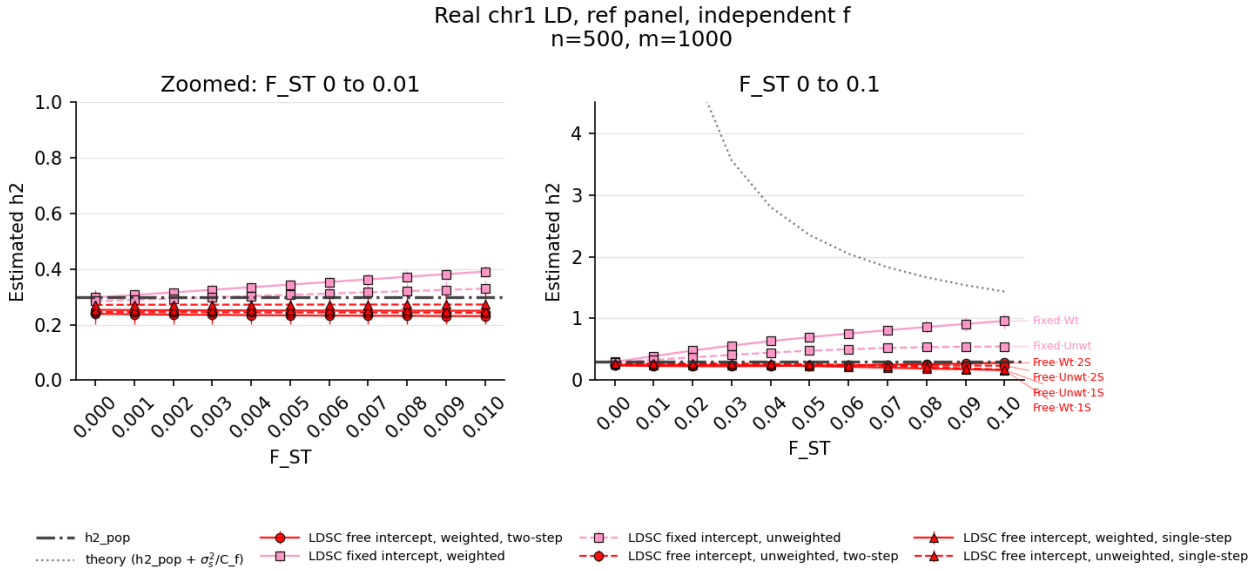


chr1, n=5000, m=10000 — ref panel, shared f

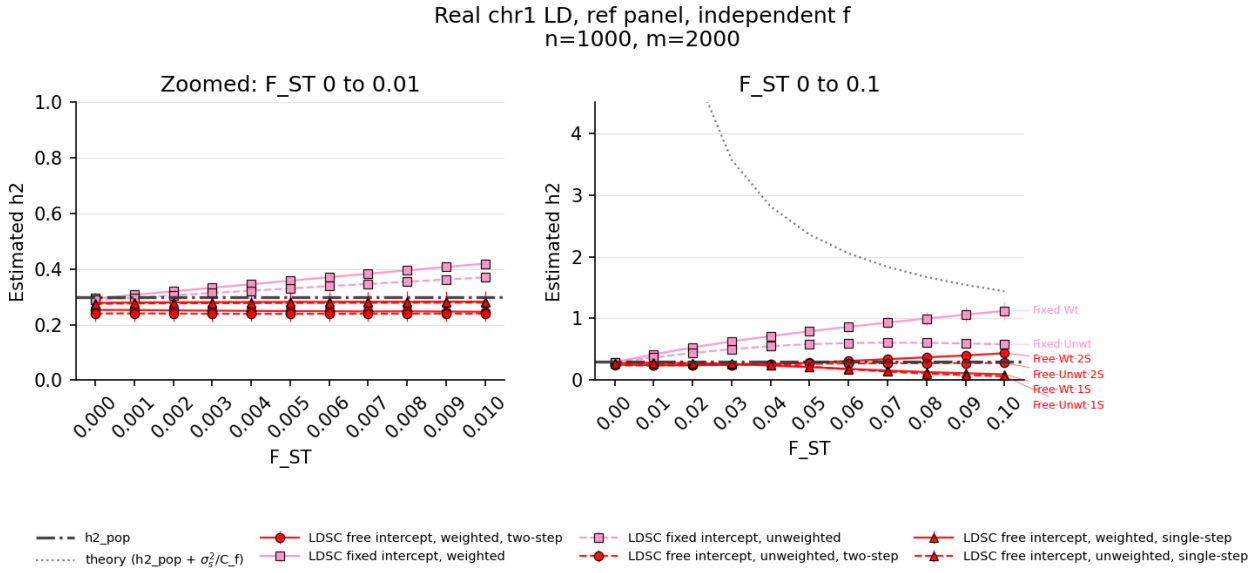


chr1, n=10000, m=20000 — ref panel, shared f (new)

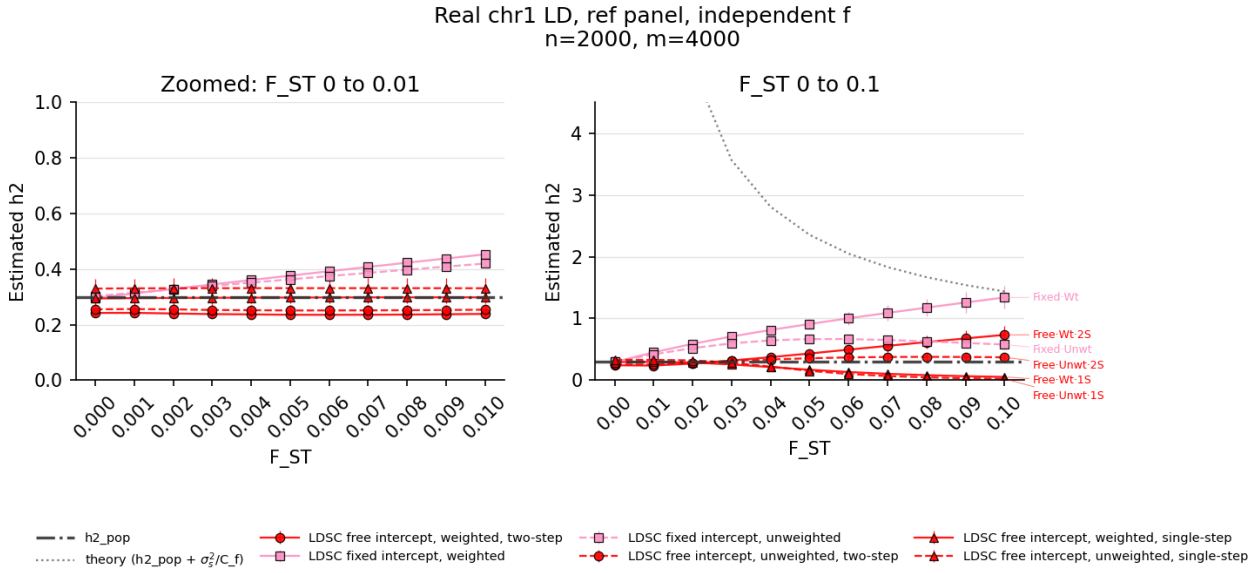
ref panel, independent f



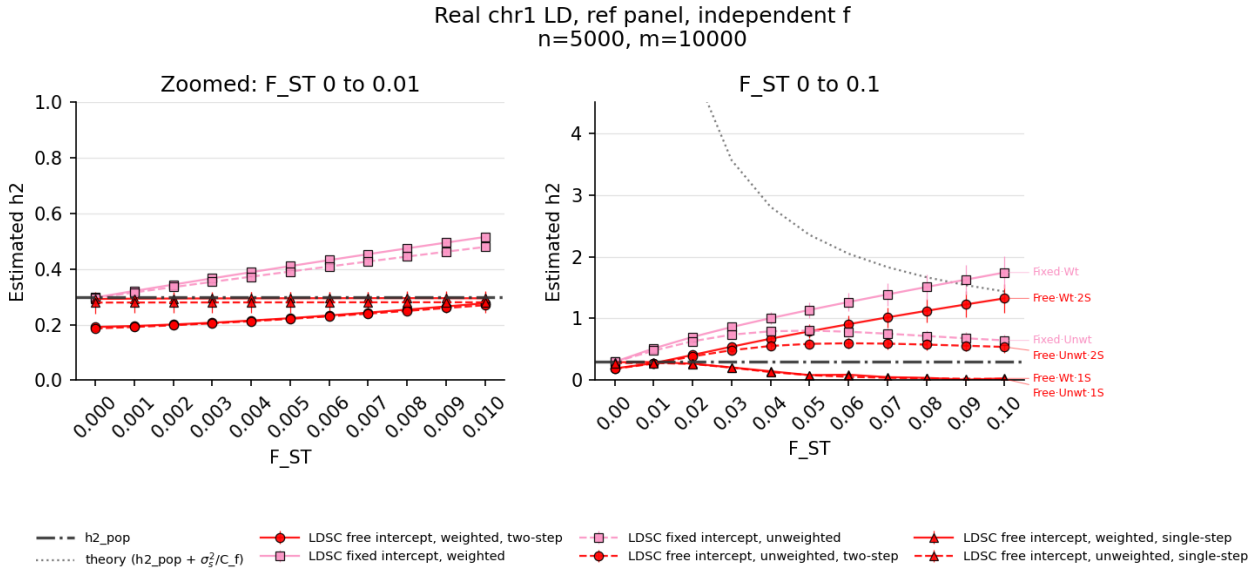
chr1, n=500, m=1000 — ref panel, independent f



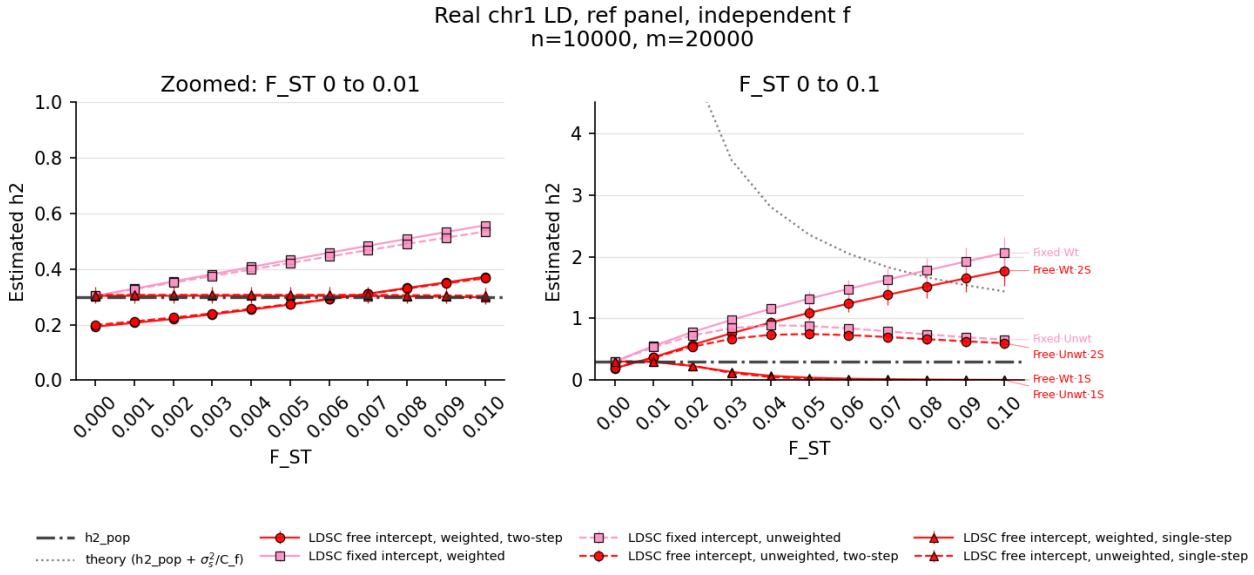
chr1, n=1000, m=2000 — ref panel, independent f



chr1, n=2000, m=4000 — ref panel, independent f



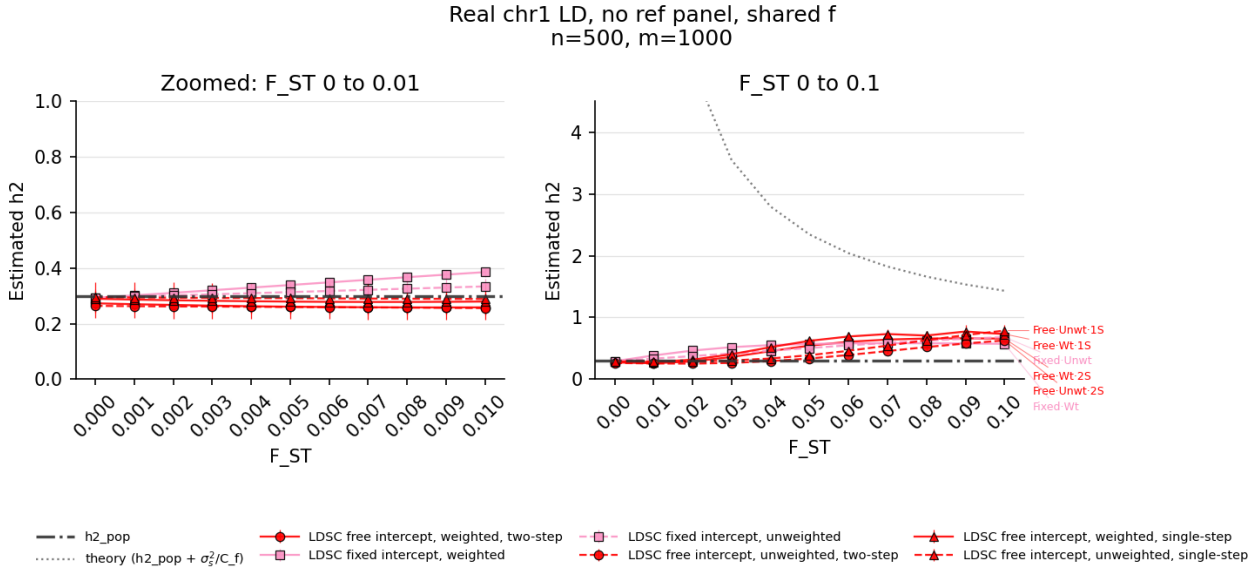
chr1, n=5000, m=10000 — ref panel, independent f



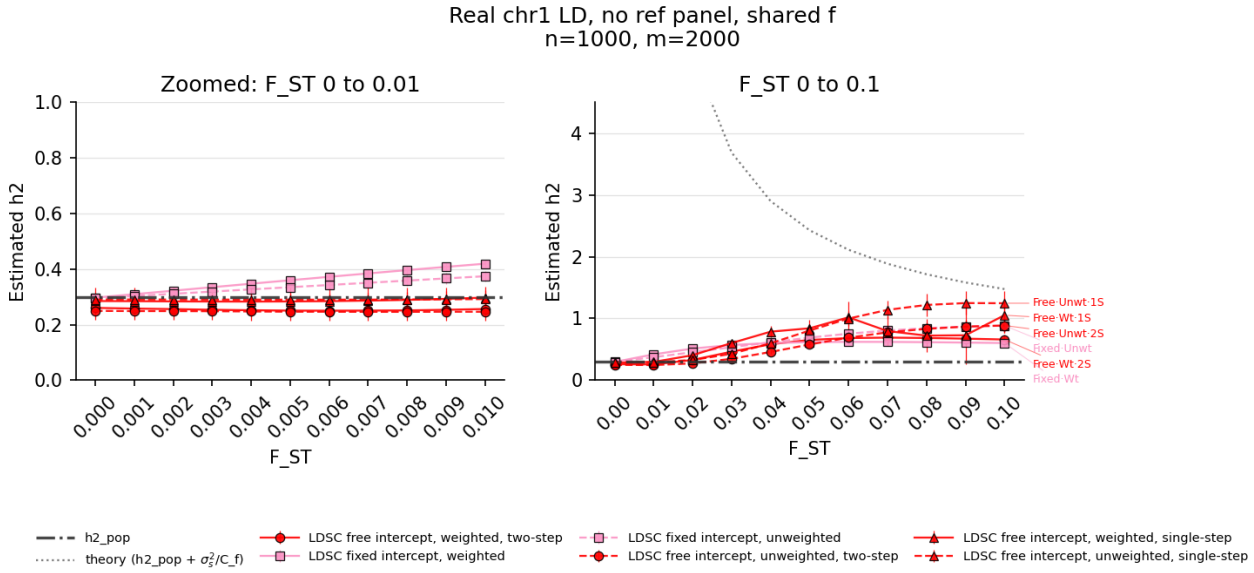
chr1, n=10000, m=20000 — ref panel, independent f (new)

No-Reference-Panel Setting (for contrast)

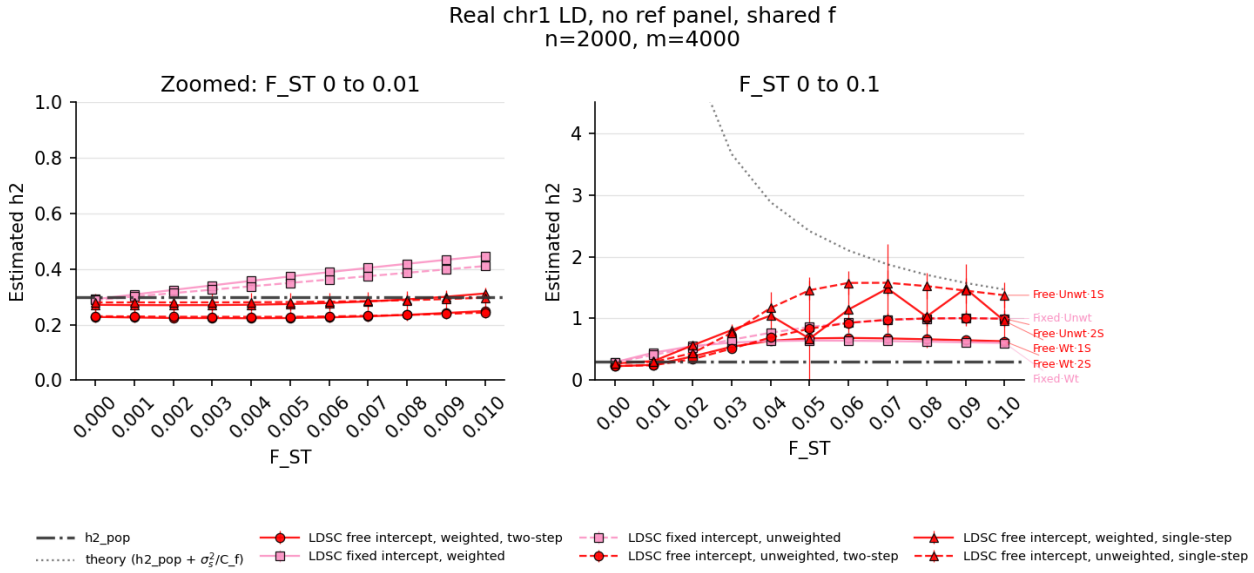
no ref panel, shared f



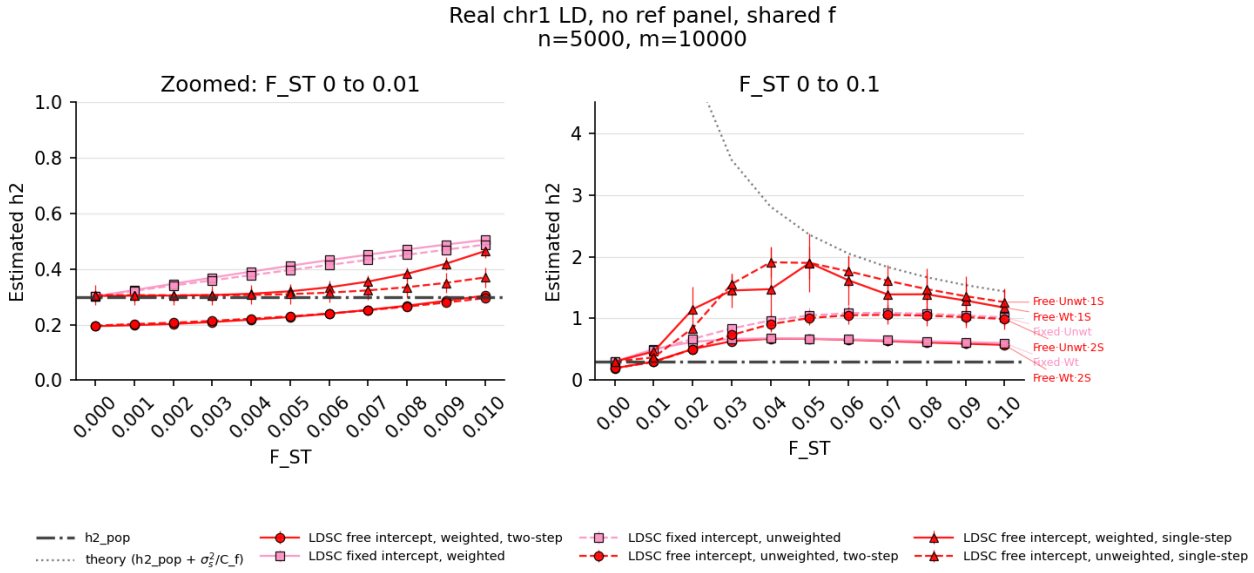
chr1, n=500, m=1000 — no ref panel, shared f



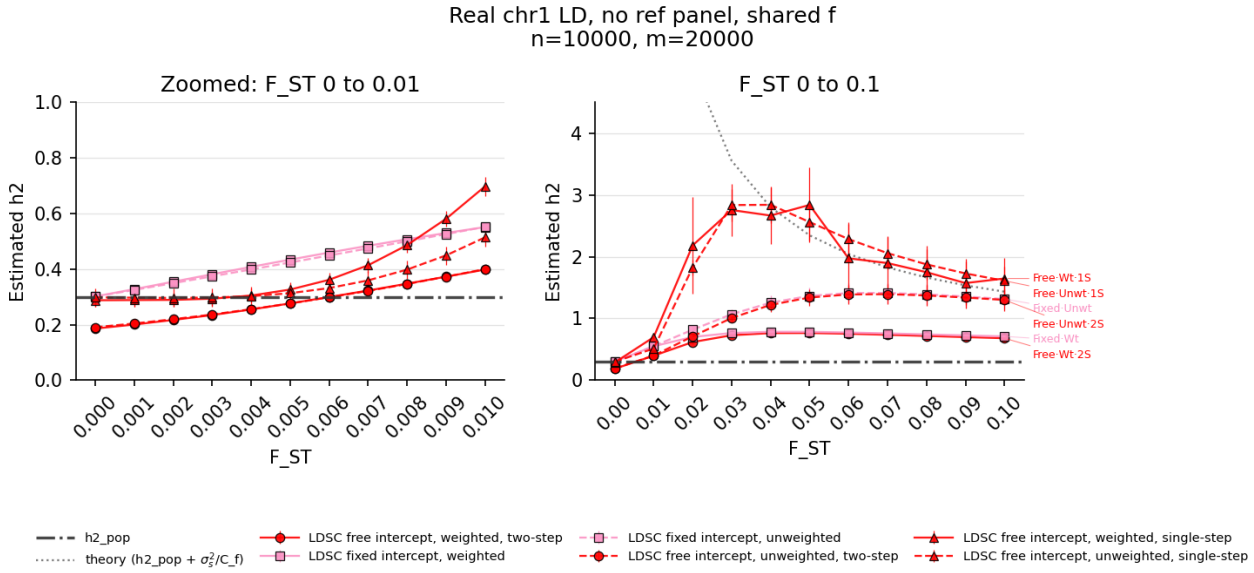
chr1, n=1000, m=2000 — no ref panel, shared f



chr1, n=2000, m=4000 — no ref panel, shared f

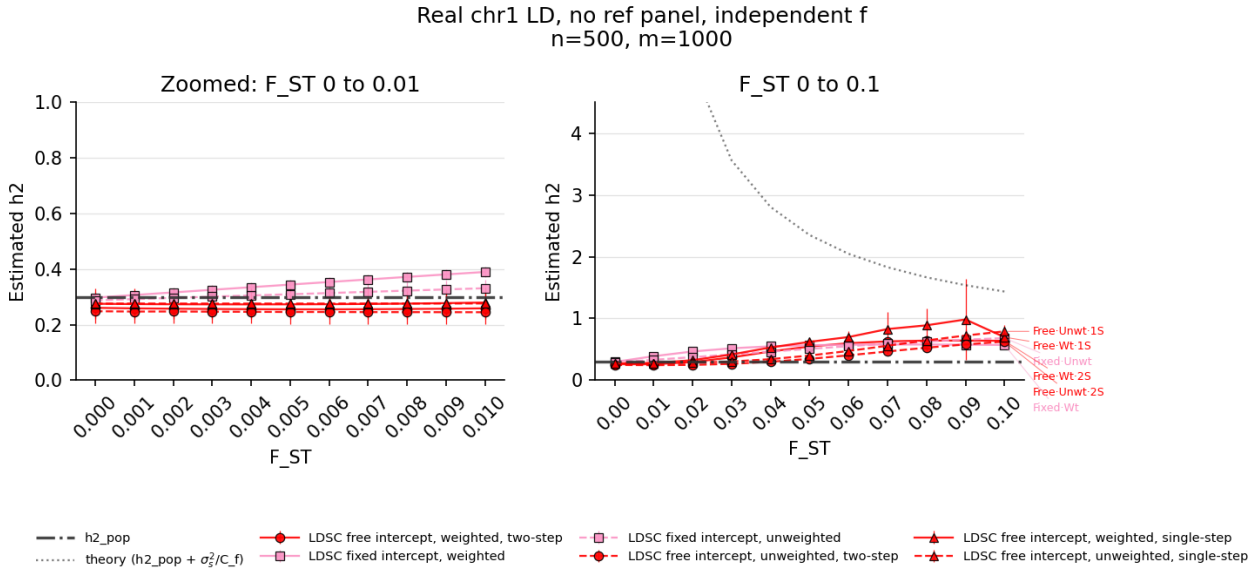


chr1, n=5000, m=10000 — no ref panel, shared f

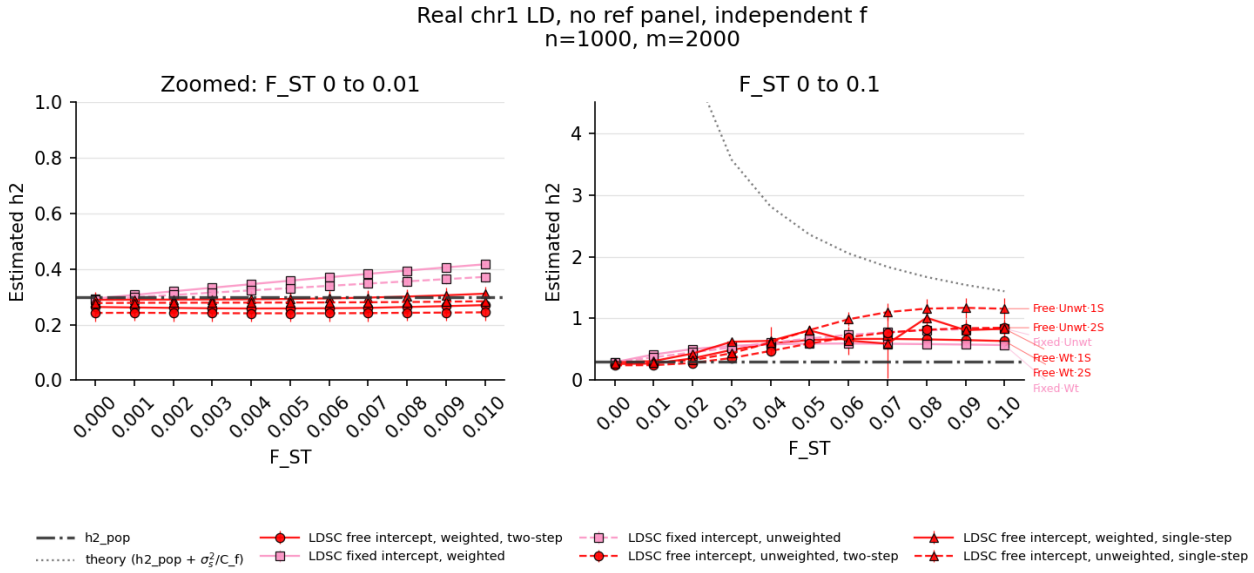


chr1, n=10000, m=20000 — no ref panel, shared f (new)

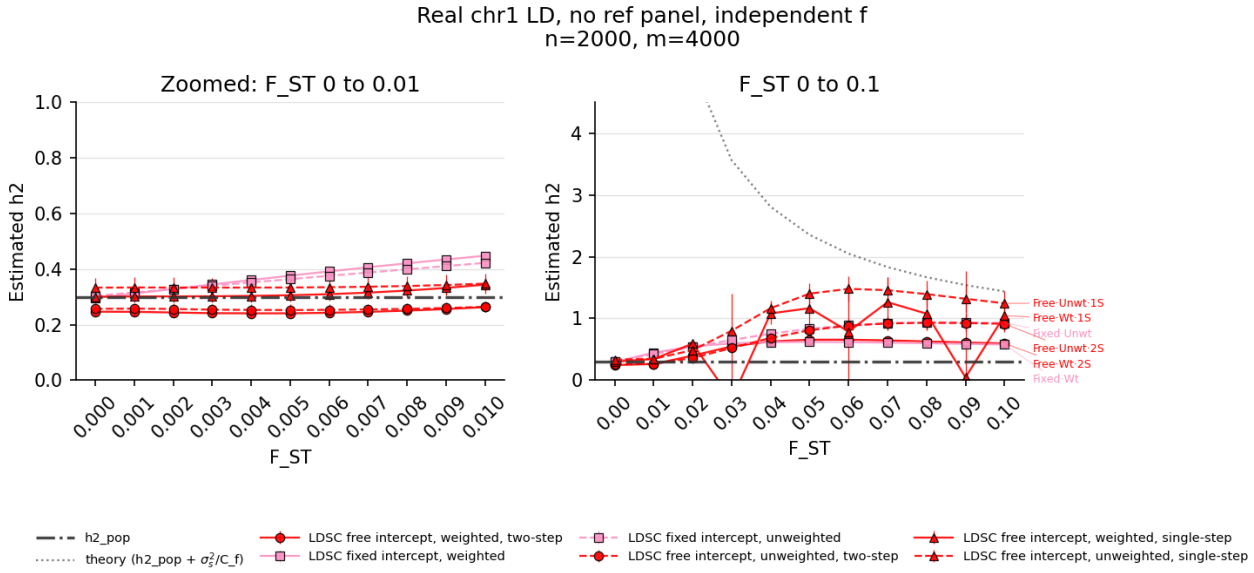
no ref panel, independent f



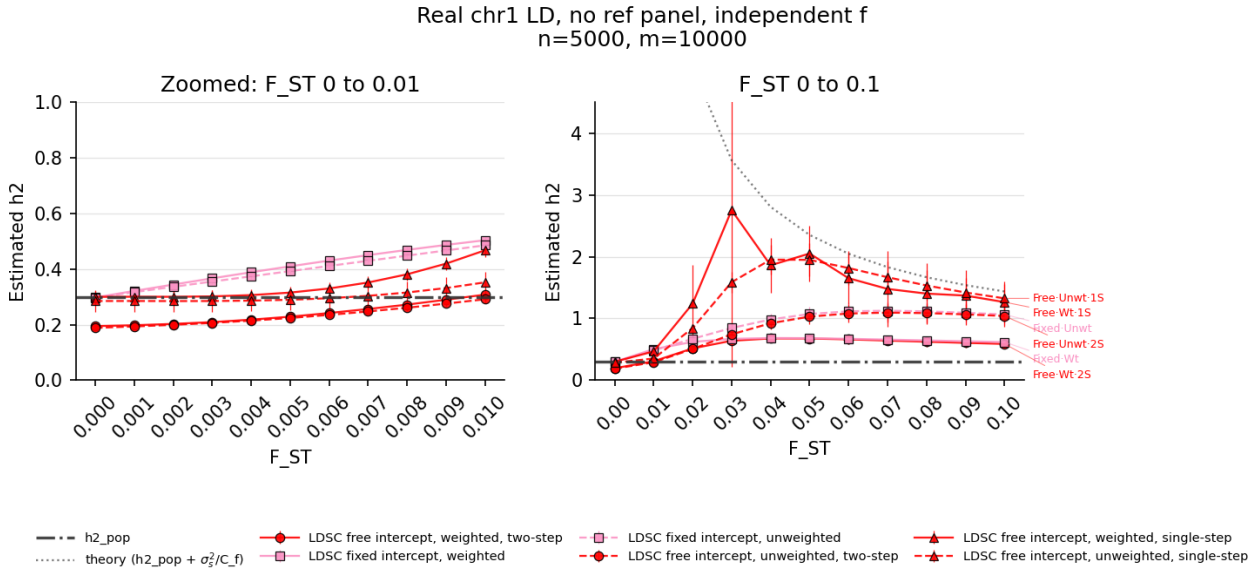
chr1, n=500, m=1000 — no ref panel, independent f



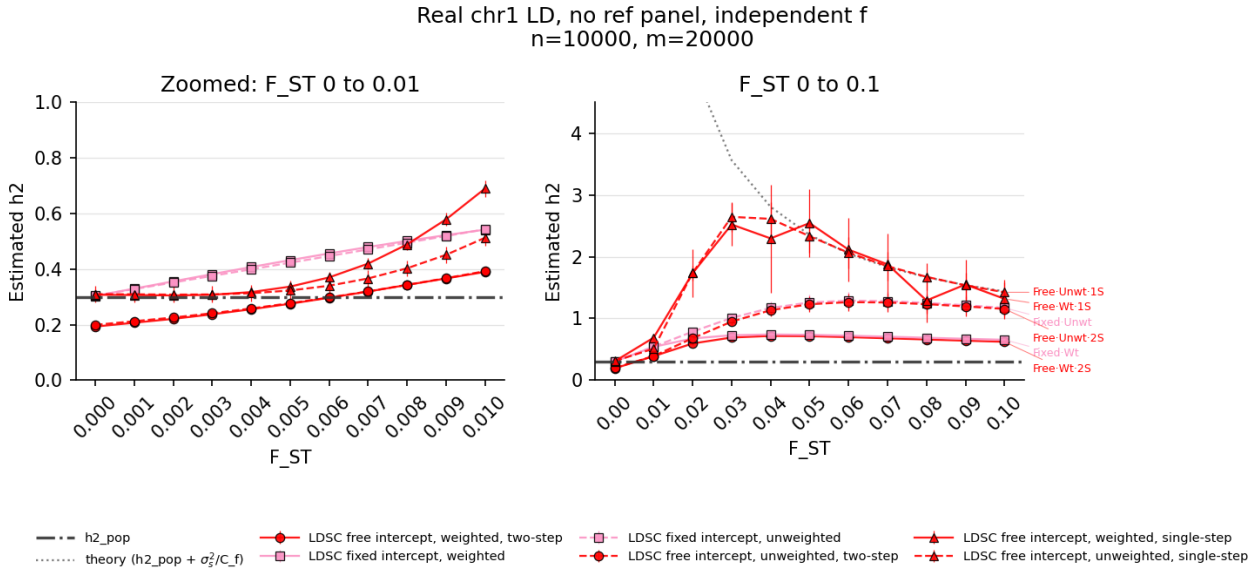
chr1, n=1000, m=2000 — no ref panel, independent f



chr1, n=2000, m=4000 — no ref panel, independent f



chr1, n=5000, m=10000 — no ref panel, independent f



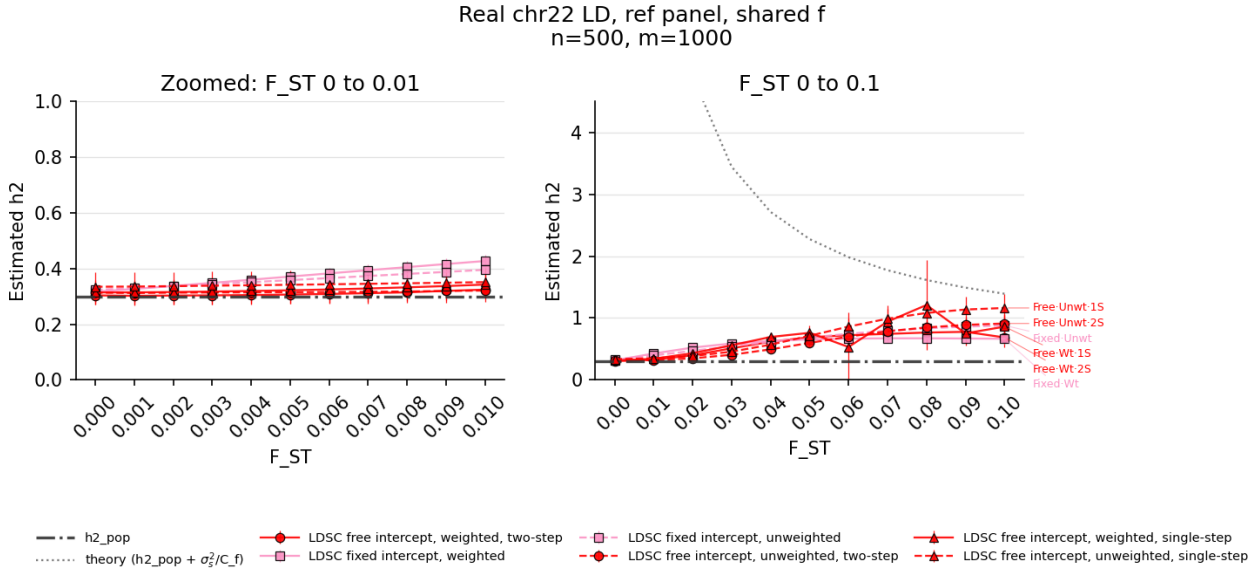
chr1, n=10000, m=20000 — no ref panel, independent f (new)

Figures: Chromosome 22

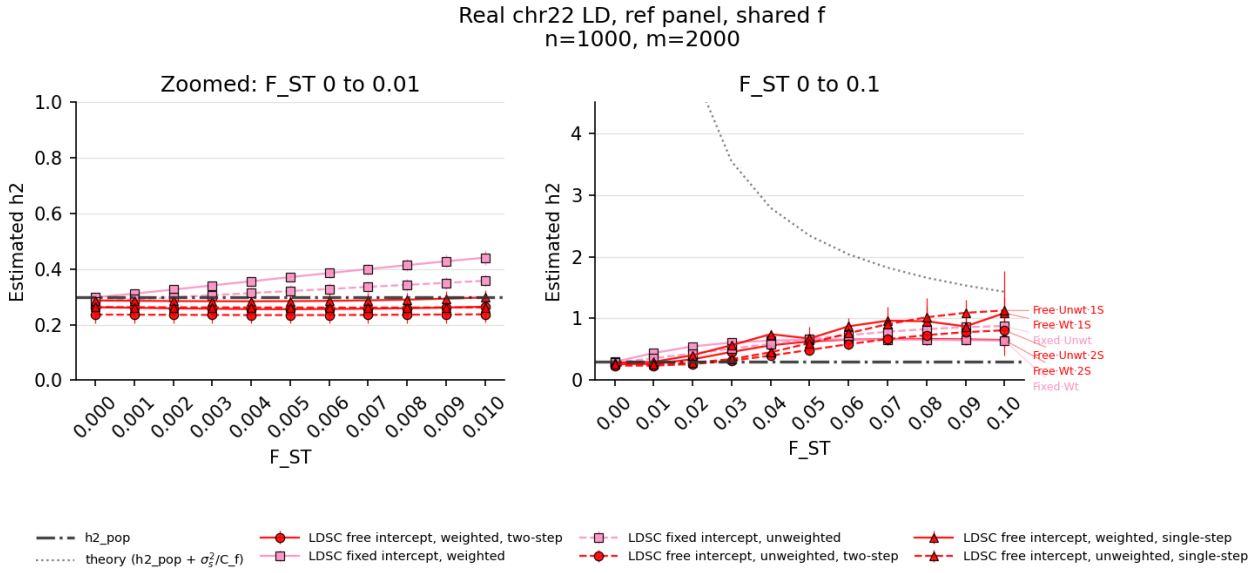
Each figure shows mean h2 (±95% CI) vs. F_ST for all six LDSC estimator variants, within one (reference-panel setting, f-sharing) combination, with the theoretical line overlaid (dotted). Color encodes intercept type (red=free intercept, pink=fixed intercept); linestyle encodes SNP weighting (solid=weighted, dashed=unweighted); marker shape encodes estimation mode (circle=two-step, triangle=single-step, square=fixed-intercept, which has no re-estimation step). Left panel zooms into F_ST 0-0.01; right panel shows F_ST 0-0.1.

Reference-Panel Setting (the main finding)

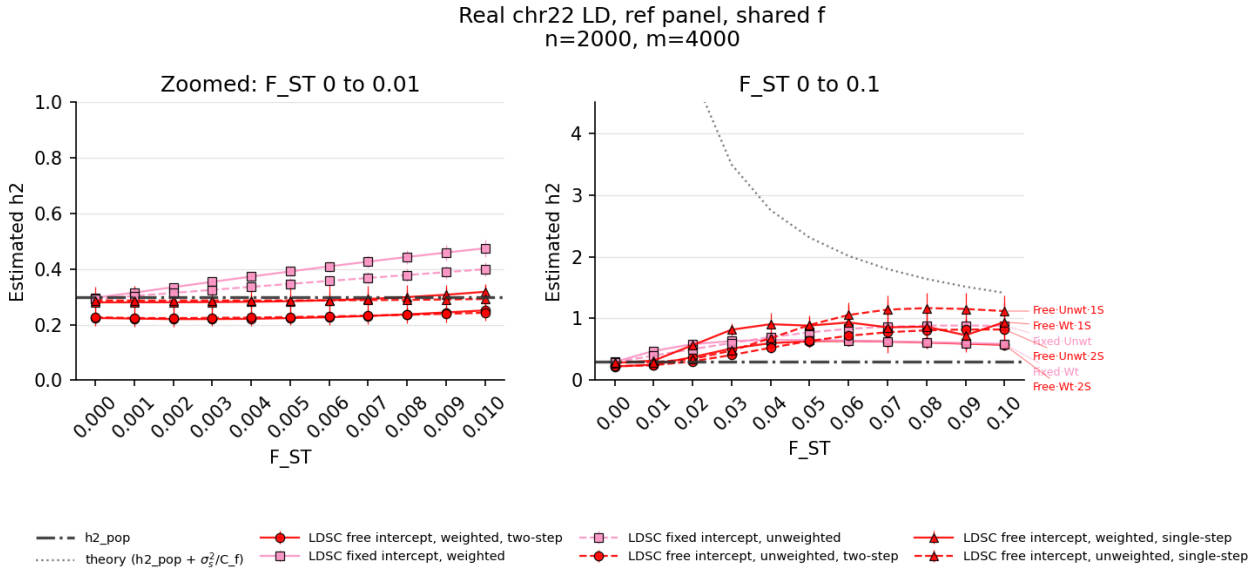
ref panel, shared f



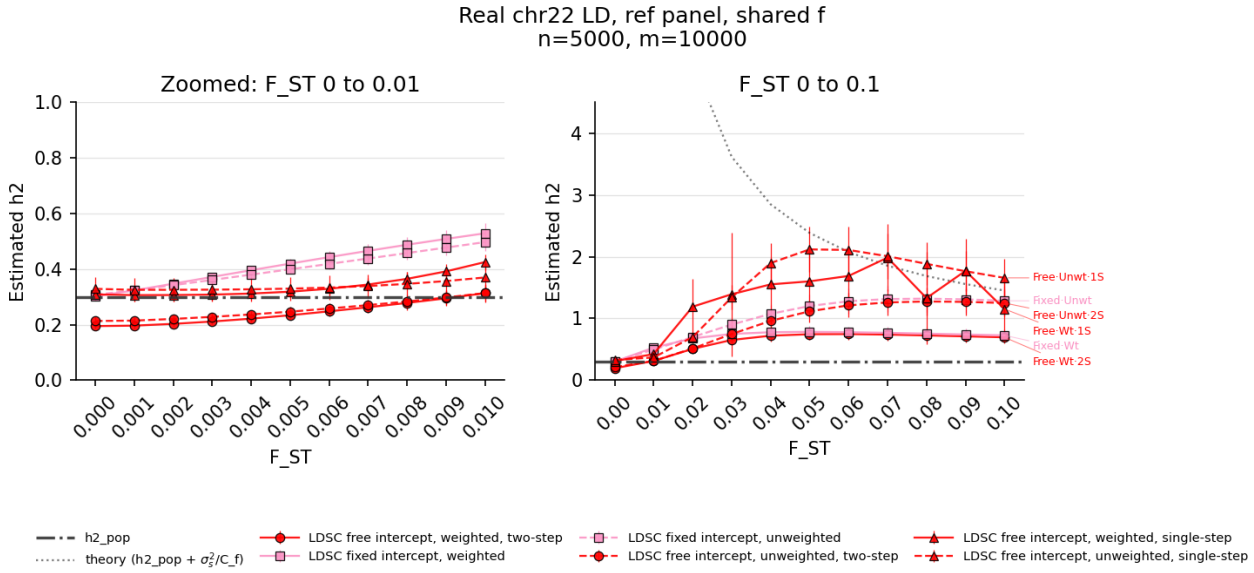
chr22, n=500, m=1000 — ref panel, shared f



chr22, n=1000, m=2000 — ref panel, shared f

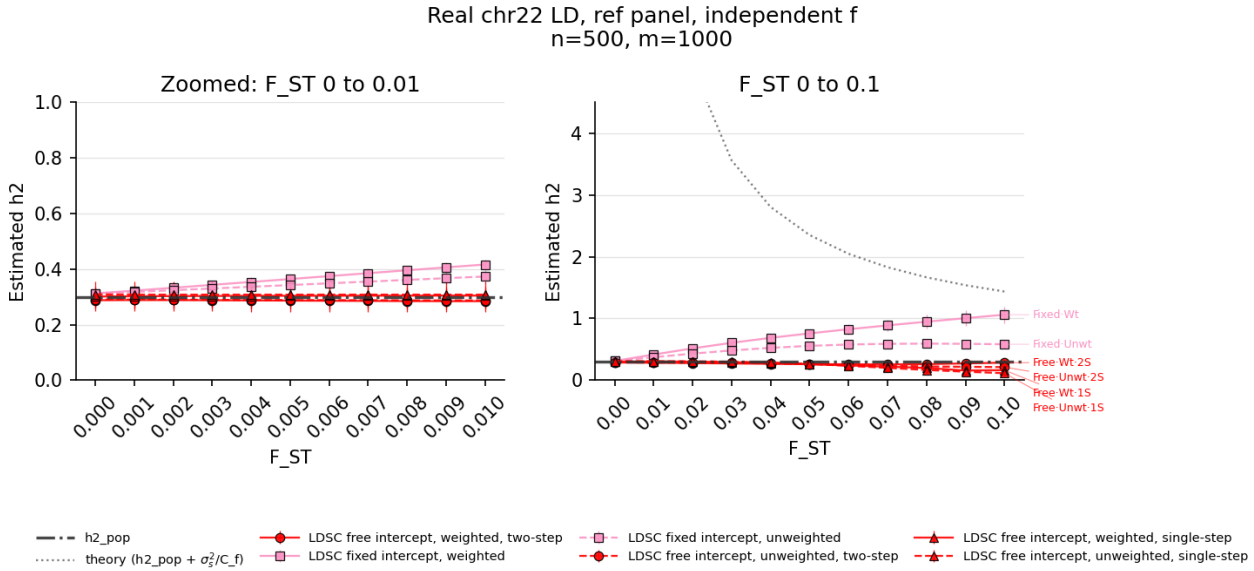


chr22, n=2000, m=4000 — ref panel, shared f

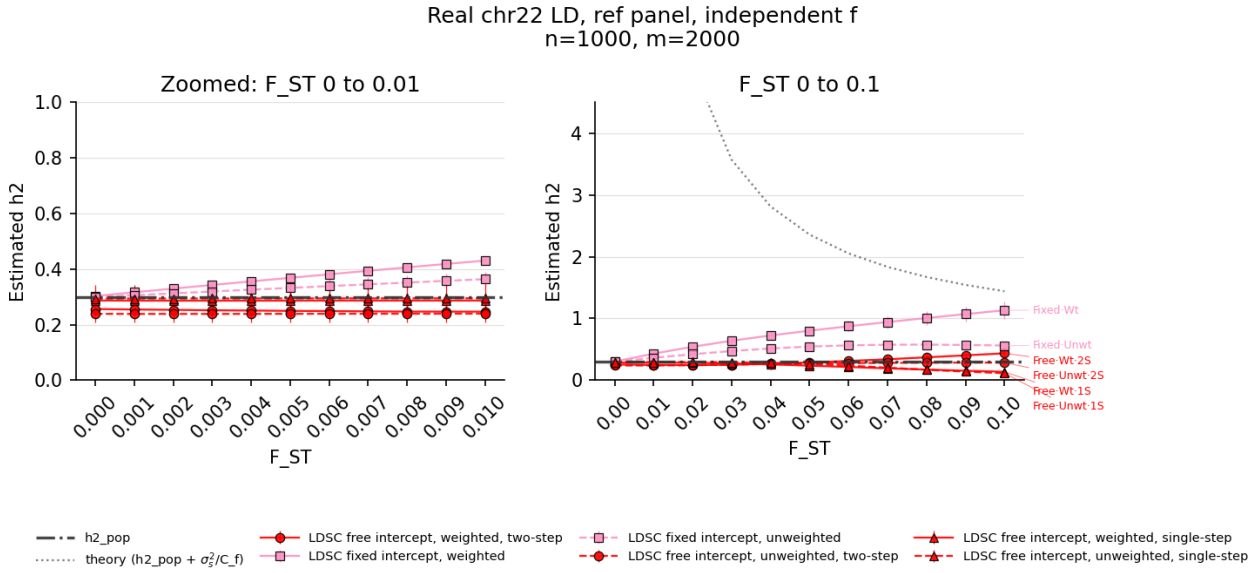


chr22, n=5000, m=10000 — ref panel, shared f

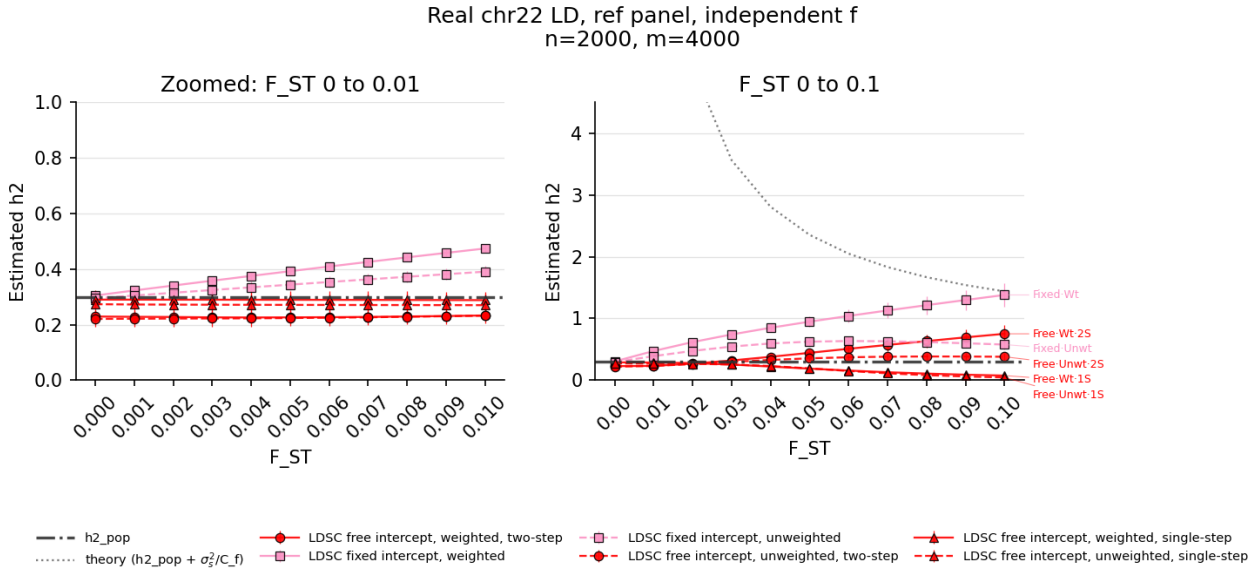
ref panel, independent f



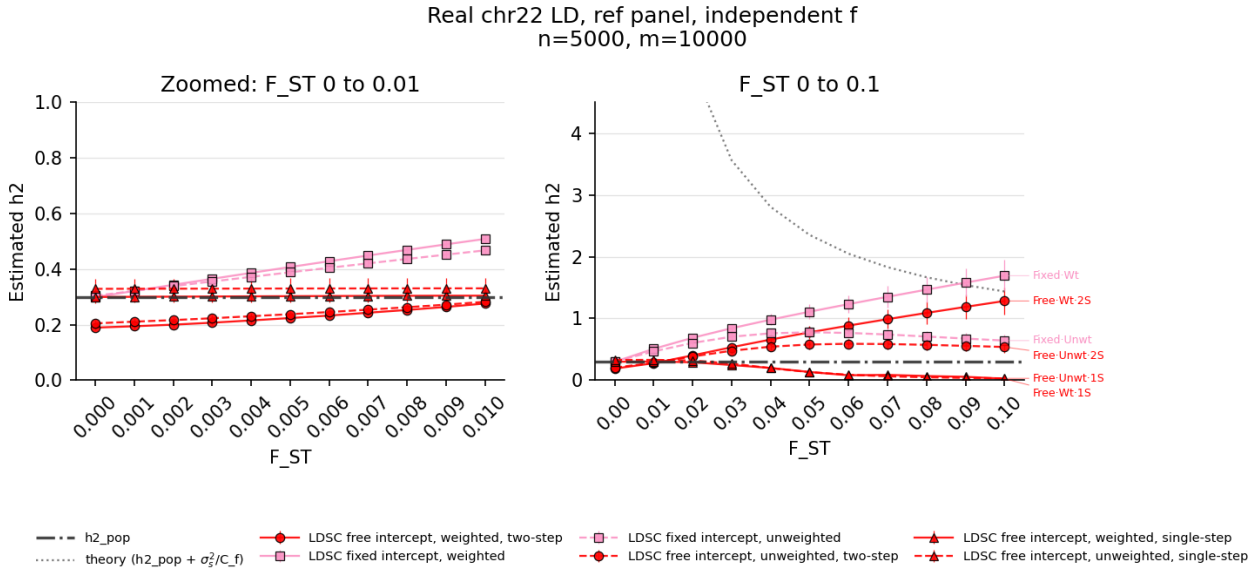
chr22, n=500, m=1000 — ref panel, independent f



chr22, n=1000, m=2000 — ref panel, independent f



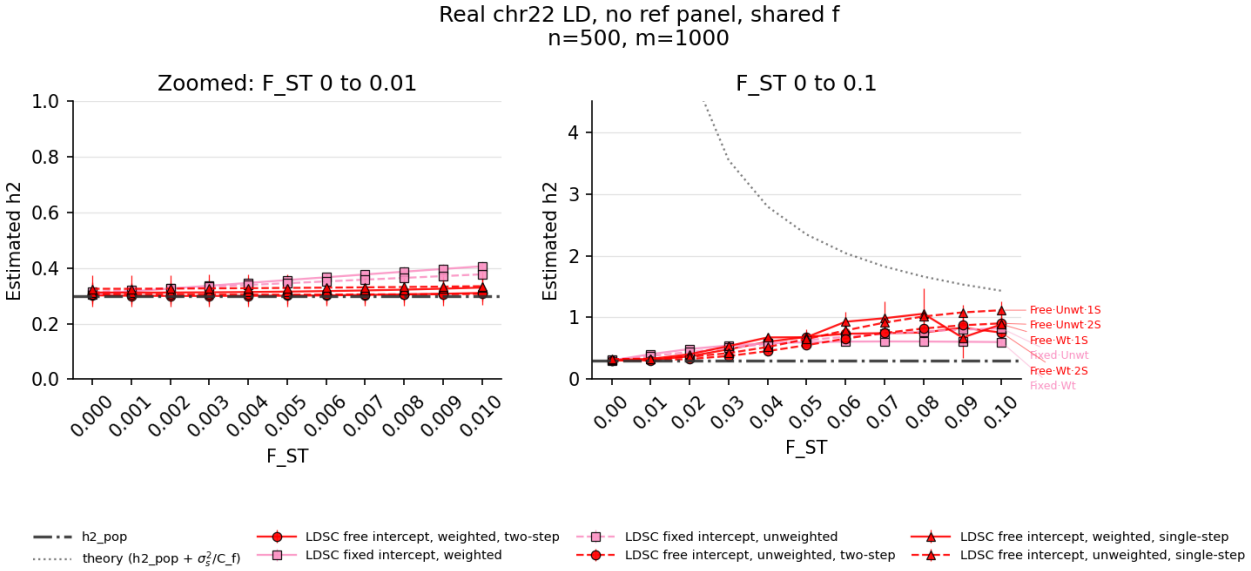
chr22, n=2000, m=4000 — ref panel, independent f



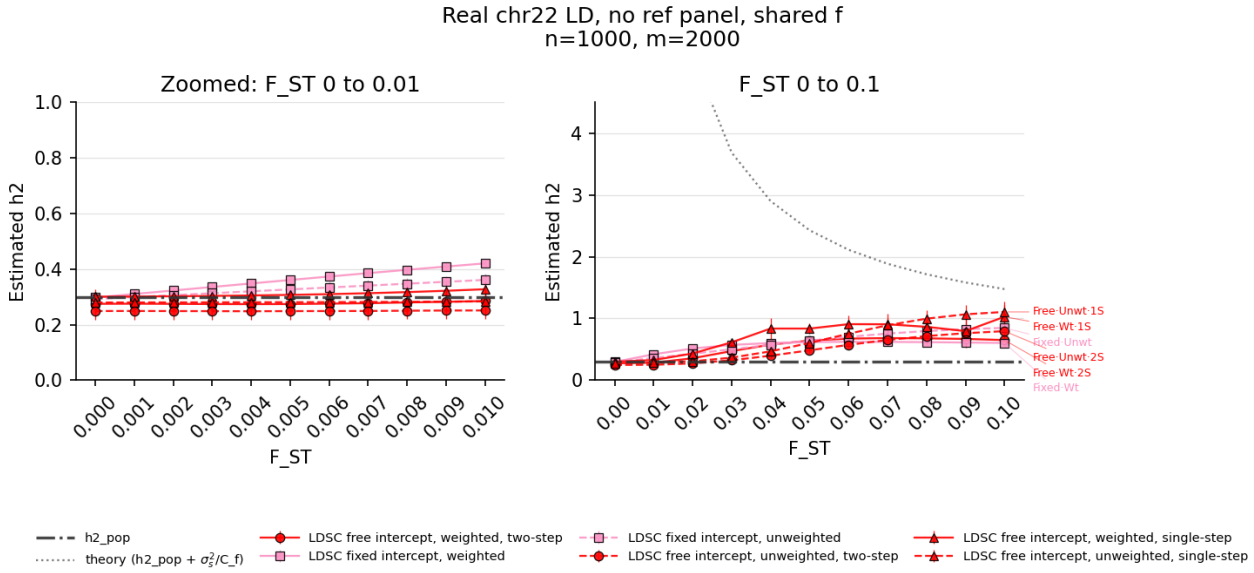
chr22, n=5000, m=10000 — ref panel, independent f

No-Reference-Panel Setting (for contrast)

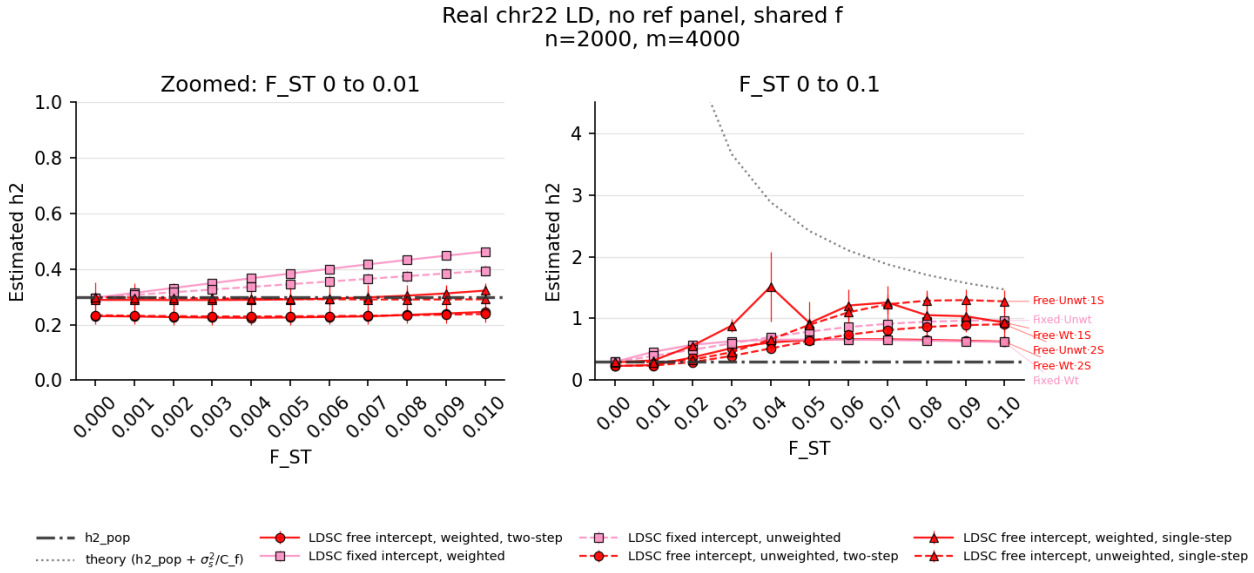
no ref panel, shared f



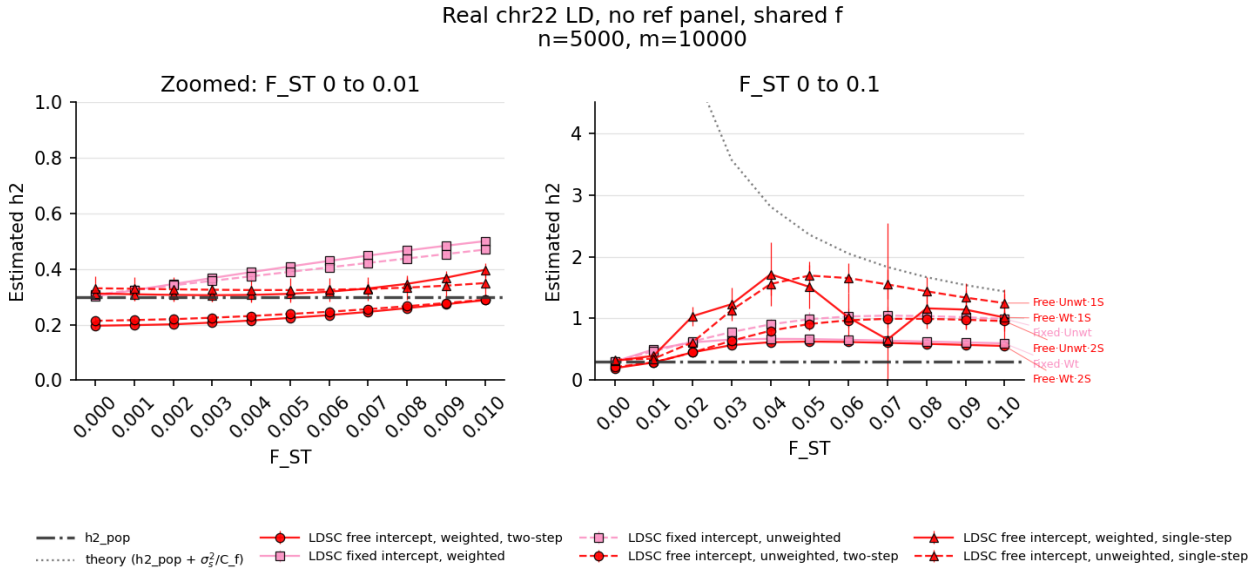
chr22, n=500, m=1000 — no ref panel, shared f



chr22, n=1000, m=2000 — no ref panel, shared f

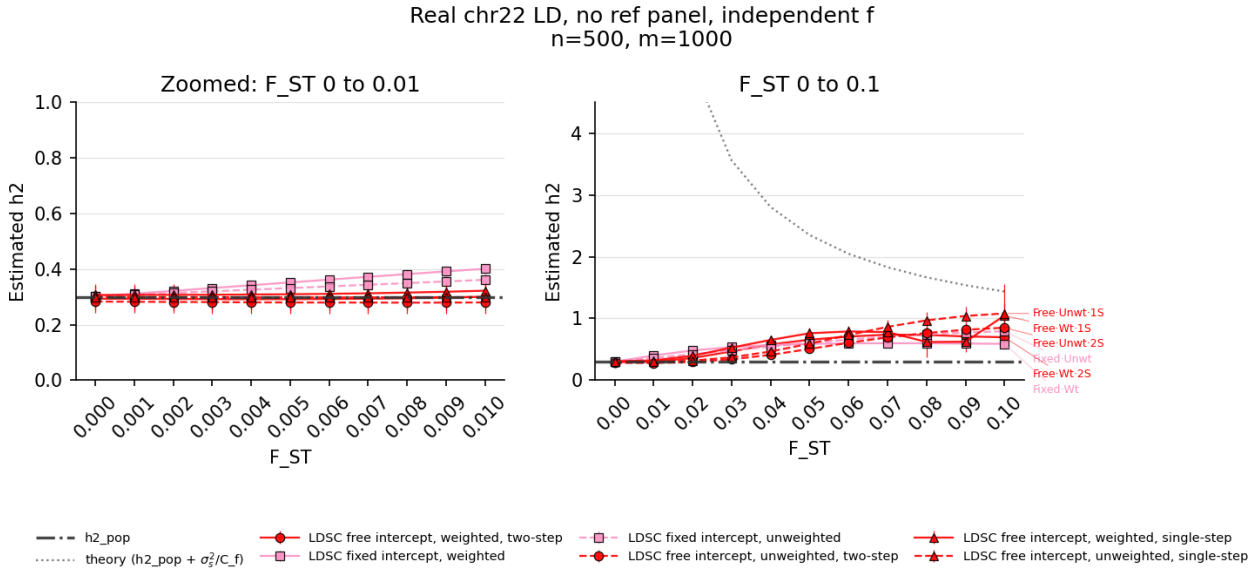


chr22, n=2000, m=4000 — no ref panel, shared f

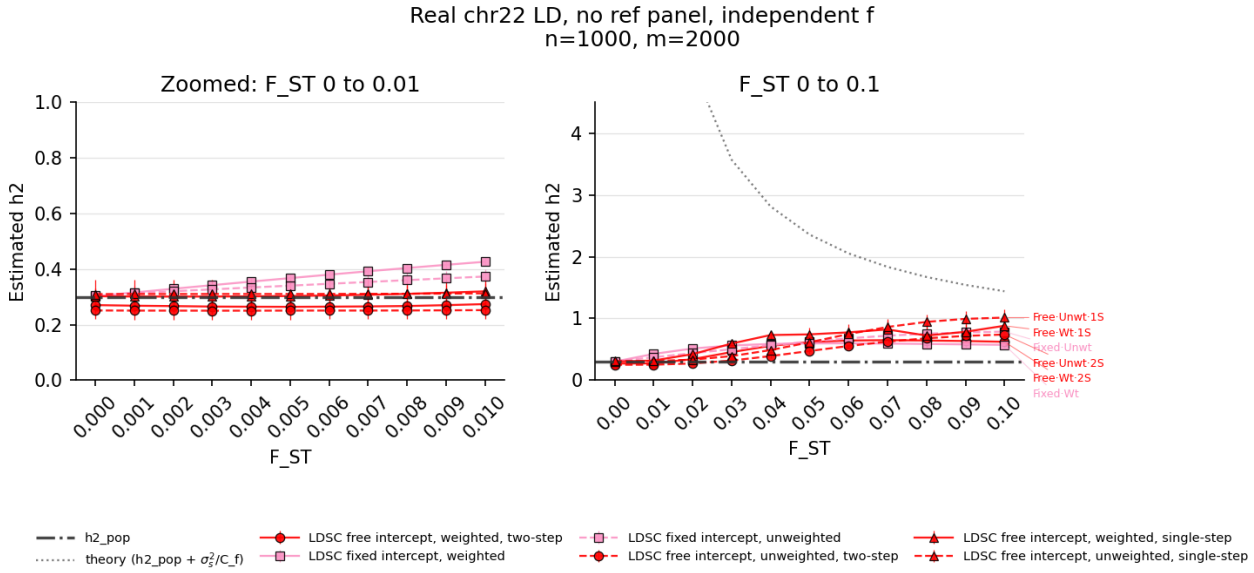


chr22, n=5000, m=10000 — no ref panel, shared f

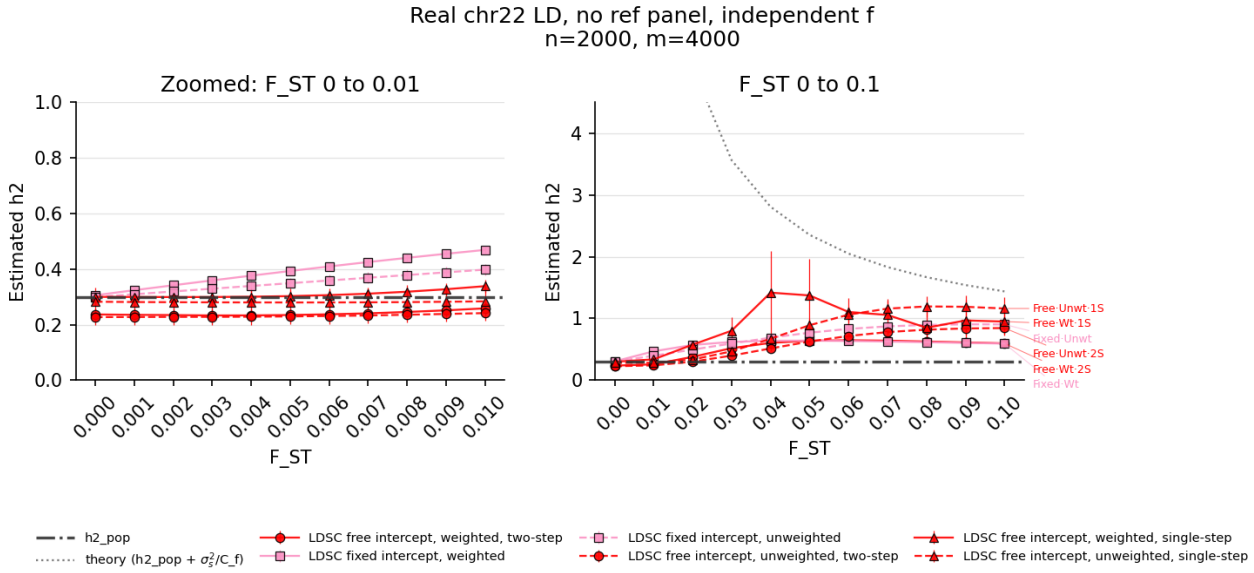
no ref panel, independent f



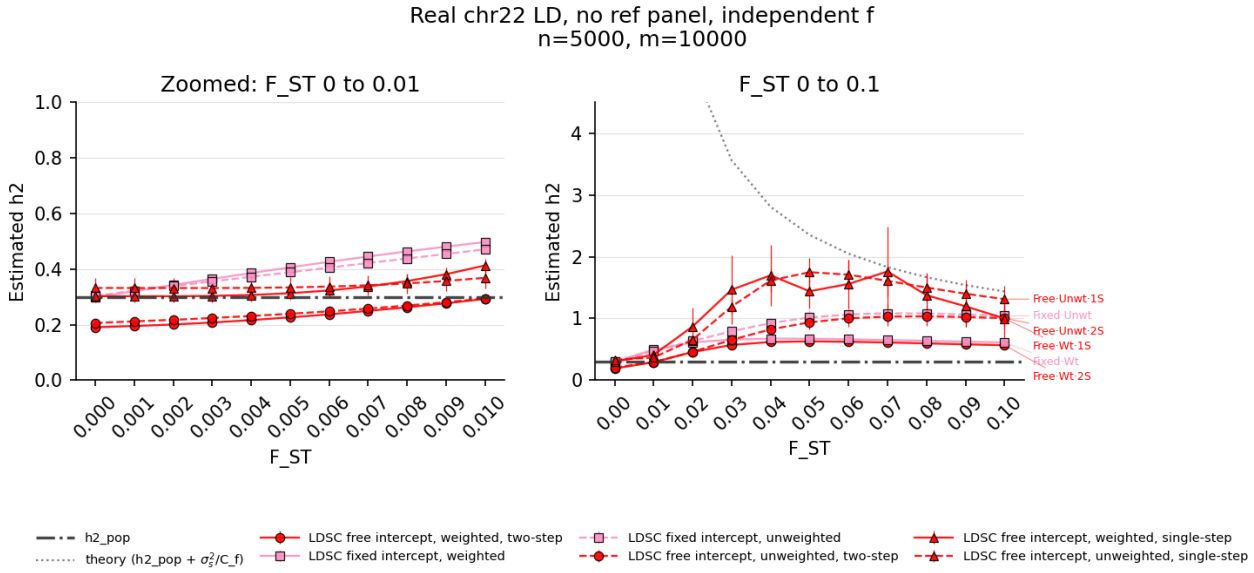
chr22, n=500, m=1000 — no ref panel, independent f



chr22, n=1000, m=2000 — no ref panel, independent f



chr22, n=2000, m=4000 — no ref panel, independent f



chr22, n=5000, m=10000 — no ref panel, independent f

Discussion

Why no-ref-panel is robust to f-sharing: when LD scores and GWASH moments are recomputed in-sample every replicate, the estimator always sees a self-consistent sample - whether that sample's own f happens to be fixed across replicates or redrawn each time is irrelevant to any single replicate's estimate. This is confirmed across all scales and both chromosomes, including the new n=10000 chr1 scale.

Why ref-panel independent-f gets worse with scale: a mismatched reference panel supplies LD scores computed from a *different* realization of population structure than the main sample's. As n and m grow, the main sample's own signal becomes more precisely estimated (tighter confidence intervals, visible in the figures), but it is being regressed against LD scores that reflect the wrong stratification pattern. The systematic bias this introduces does not shrink with more data - sample size reduces variance, not the bias from using a structurally mismatched reference panel - so the gap between shared-f and independent-f widens as the estimates themselves become more precise. The n=10000/m=20000 chr1 figure shows this same widening starting within the 0-0.1 range already (shared f saturating while independent f keeps climbing toward F_ST=0.1), even without re-running the older 0.2-0.5 grid at this scale.

Inflation vs. collapse: this contrasts with the earlier AR1-based reference-panel finding, where an f-mismatched reference panel caused the free-intercept estimator to collapse toward zero rather than inflate. The sign of the failure mode appears to depend on the LD structure used (synthetic AR1 vs. real chromosome LD), which is itself a notable result: it is not safe to assume the direction of bias from a mismatched reference panel without knowing the underlying LD structure.

chr1 vs. chr22: the near-identical gap trajectories (within ~0.1-0.15 of each other at every shared scale) across two chromosomes of very different size (106,136 vs. 18,613 SNPs) and gene density suggest this is primarily a sample-size effect of the simulation design, not an artifact of a particular chromosome's LD architecture.

Practical implication: these results reinforce that a reference panel is only useful for this kind of heritability estimation if its stratification structure genuinely matches the target sample's - and that mismatches do not wash out with larger studies. If anything, larger reference-panel-based studies have more to lose from an f-mismatch than smaller ones.

Appendix: Data and Reproducibility

All 848 simulation result CSVs (4 scenarios × chr1 scales n=500-10000 + chr22 scales n=500-5000, 100 replicates each) are saved under data/python_realistic_* (chr1) and data/python_realistic_*_chr22 (chr22) in the project repository, and are not regenerated by the plotting scripts - replotting or re-analysis can be done directly from these files.

Driver scripts (python/):

- do_realistic_sim_Sam_noref_sharedf[_chr22].py, do_realistic_sim_Sam_noref_indepf[_chr22].py
- do_realistic_sim_Sam_ref_fixedf[_chr22].py, do_realistic_sim_Sam_ref_indepf[_chr22].py
- make_ref_panel_realistic[_chr22].py
- plot_realistic_sharedf_vs_indepf[_chr22].py <n> (regenerates figures from saved CSVs only)
- generate_realistic_sharedf_vs_indepf_report.py (regenerates this report PDF from saved CSVs only)

Each driver script accepts F_ST and an optional sample size n (m = 2n, default n=500) as command-line arguments. n=10000/m=20000 chr1 noref_indepf results were generated on a separate Unix machine (to avoid repeated Windows crashes during a multi-day run) and verified numerically equivalent to this machine's output for an identical combo before being merged in (see python/compare_platform_outputs.py and python/logs/platform_benchmark/).